VLA-Corrector

Lightweight Detect-and-Correct Inference for Adaptive Action Horizon

Embodied AI · Vision-Language-Action · Action Correction · ~40M Corrector

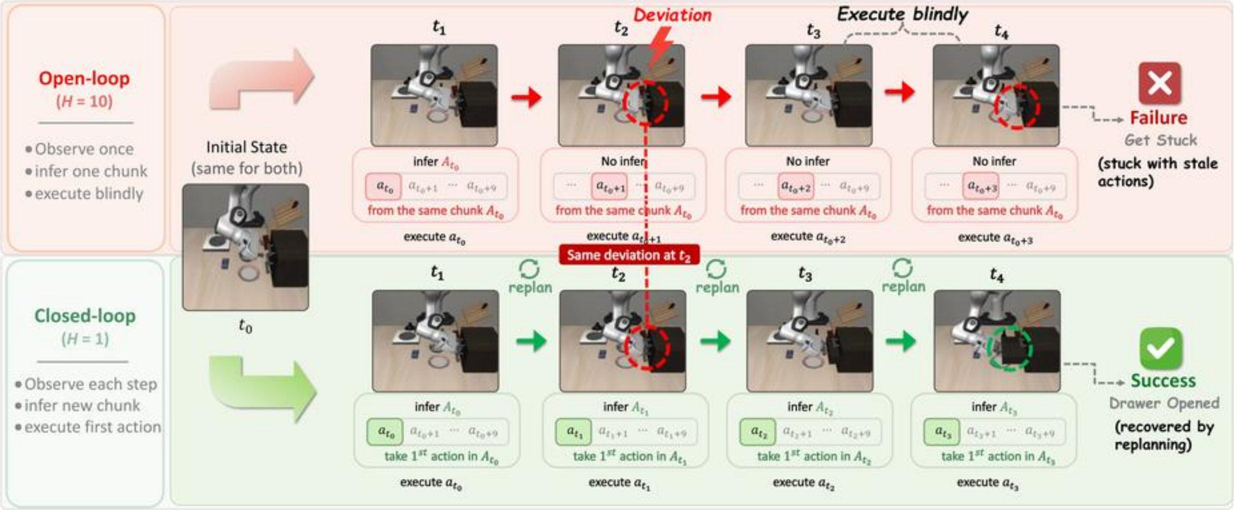
Motivation

Can a frozen VLA keep long-horizon efficiency while recovering from execution drift before stale actions accumulate?

Code: github.com/ZJU-OmniAI/vla-corrector · arXiv: arxiv.org/abs/2607.01804

01 MOTIVATION

Open-loop VLA execution leaves blind spots.



Long action chunks

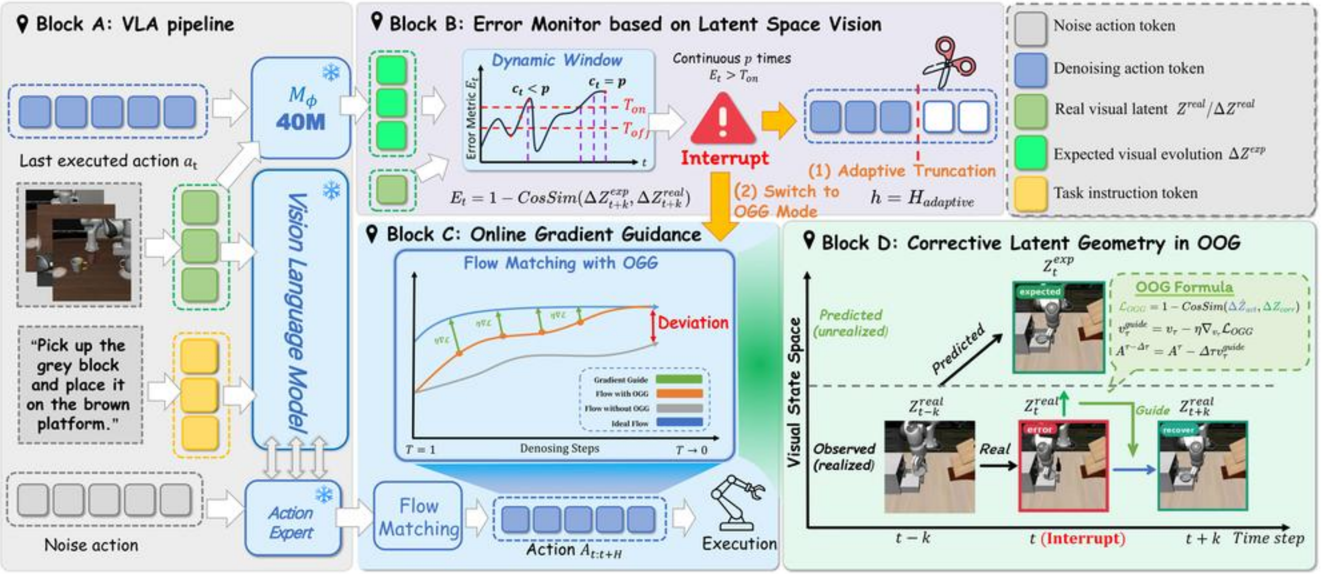
Efficient VLA calls, but queued actions can become stale after drift.

Per-step replanning

More reactive, but expensive for large VLA policies.

02 METHOD

Detect drift, truncate stale chunks, guide recovery.

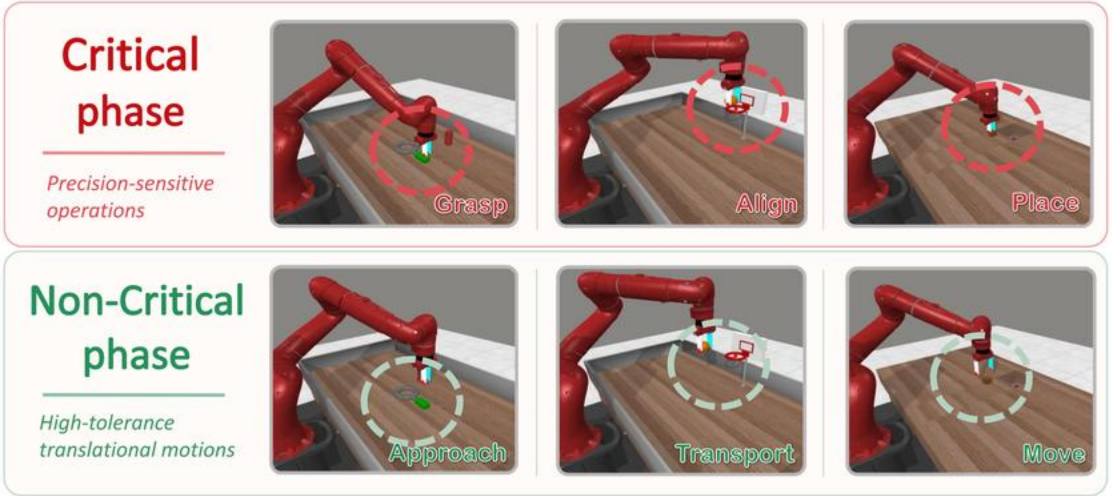


01 **Frozen VLA backbone with an external latent dynamics corrector.**

02 **Latent-space Vision Monitor detects persistent visual dynamics mismatch.**

03 **Online Gradient Guidance is applied only to the recovery query.**

A small external model instead of full VLA retraining.



Trainable component

~40M

Residual MLP corrector reported at roughly 38--42M parameters.

Policy backbone

Frozen

The base VLA is not fully retrained for correction.

Action horizon

Adaptive

Stable chunks continue; stale chunks are interrupted.

04 RESULTS

Reported gains across simulation and real robots.

MetaWorld / PI0.5

+15.65 pts

48.70% -> 64.35% avg. success

LIBERO / PI0.5

+3.80 pts

94.00% -> 97.80% few-shot success

AgileX PiPER

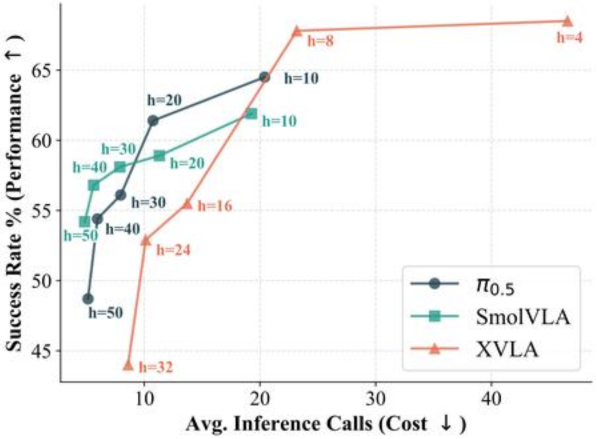
+17.7 pts

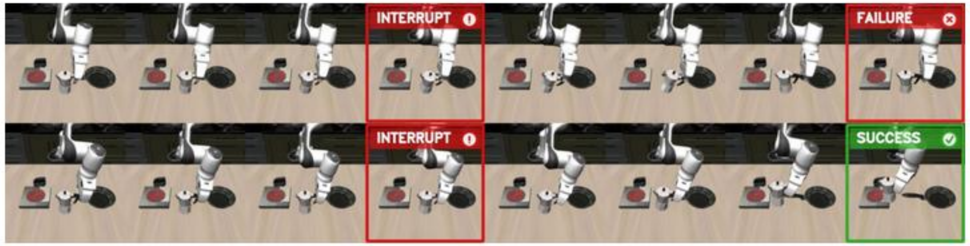
55.6% -> 73.3% real-world success

Critical phases

83.7%

Truncations in manually labeled critical phases





Qualitative recovery: truncate stale chunk, replan with OGG, complete the task.

Perturbation demos are kept as silent project-page clips.

Drawer alignment

Human moves the drawer during execution.

Block to blue bowl

Target bowl is shifted mid-task.

Block to white bowl

Object/target perturbation during execution.

The presentation uses static thumbnails; the project page still hosts the three silent demo videos.

VLA-Corrector

Resources

Code	https://github.com/ZJU-OmniAI/vla-corrector
Project page	https://zju-omniai.github.io/vla-corrector/
Paper	https://arxiv.org/abs/2607.01804
arXiv	https://arxiv.org/abs/2607.01804
Editable slides	docs/assets/presentation/vla_corrector_presentation.pptx